📘 PERIODIC TABLE (ADV → OLYMPIAD LEVEL)

1. Effective Nuclear Charge (Zeff) – CORE

Zeff = actual nuclear pull on outer electron

Key idea:

Across period → Zeff sharply increases

Down group → Zeff almost constant (shielding cancel karta hai)

👉 Insight: Isliye size ↓ hota hai across, but down me itna drastic change nahi hota

2. Atomic Radius – DEEP LOGIC

Factors:

Zeff

Number of shells

Shielding

Order (important traps):

cation < atom < anion

isoelectronic series:

more protons → smaller size

Example: $O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

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3. Ionization Energy (IE) – OLYMPIAD VIEW

Definition: energy to remove electron

Factors:

Zeff

size

penetration (s > p > d > f)

Important exceptions (must know):

Be > B (filled vs partially filled)

N > O (half-filled stability)

👉 Advanced insight:

d and f electrons poor shielding → transition elements me irregular trend

4. Electron Affinity (EA) – TRICKY AREA

Definition: energy released when electron added

Key trends:

Generally increases across

But irregular

Important exception:

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Cl > F

(reason: F me small size → repulsion)

5. Electronegativity (EN)

Depends on:

Zeff

size

Advanced rule:

Hybridization effect: sp > sp2 > sp3

👉 More s-character → nucleus ke close → higher EN

6. Diagonal Relationship (OLYMPIAD FAVORITE)

Li ~ Mg

Be ~ Al

Reason:

similar charge density

similar polarising power

7. Inert Pair Effect

Heavy p-block elements me ns2 electrons inert ho jate hain

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Example: Tl^+ more stable than Tl^{3+}

👉 Cause:

poor shielding by d and f electrons

8. Variable Oxidation States

Transition metals: due to (n-1)d and ns electrons

Example: Fe^{2+}, Fe^{3+}

🔥 PERIODIC TABLE FINAL INSIGHT

Properties = Zeff + shielding + orbital penetration

Exceptions hi questions banate hain (ADV + OLYMPIAD)

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